PESTCAST

Pathway Pest Risk Intelligence

Geospatial surveillance allocation for USDA APHIS PPQ

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The Allocation Problem

Fruit fly establishment costs U.S. agriculture an estimated **$1.4 billion per year** in potential losses. APHIS surveils every port of entry to intercept detections before populations establish — but resources are finite.

With **250 million international arrivals annually** and **3,200 U.S. counties**, the operational search space is enormous. Program managers must decide where to deploy inspector-hours, where to set traps, and which counties to prioritize — every quarter.

⚠ Today, allocation relies primarily on intuition and prior-year interception counts. No quantitative tool ranks risk at county × month × species granularity and translates that ranking into operational guidance.

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What PESTCAST is:

PESTCAST is a forecast-and-allocate decision-support tool for fruit fly surveillance — built on eight public datasets and calibrated against APHIS detection records.

Forecasts

Predicts fruit fly detection risk at county × month × species grain for the next 12 months. The next **90 days** are marked as the operational window.

Recommends

Ranks where additional inspector-hours catch the most pests, accounting for **diminishing returns** at already-saturated cells.

Justifies

Every prediction is **fully traceable** through fitted model coefficients, with credible intervals and an honest residuals view for audit readiness.

Six modeled fruit fly species · 50 US states · 3,200 counties · CY2026 forecast horizon

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How PESTCAST is built

Eight public datasets · four stages · ~3 minutes end-to-end · re-runs monthly

ASSEMBLE

- BTS T-100 (11 yr passenger + freight)
- USDA FAS GATS (host-commodity imports)
- EPPO presence (6 fruit fly species)
- APHIS PDR (52 detection labels)
- → one canonical Parquet at (origin × port × month) grain

ENRICH

- Network-structure features per state (origin diversity, Shannon entropy, infested-share)
- Climate suitability per county (WorldClim 10-arcmin · longest cyclic run of favorable months per species envelope · zonal stats over 3,221 counties)

FIT & PREDICT

- Poisson GLM fitted at state-month-species grain (log volumes, Fourier seasonality, network features, species FX)
- Pseudo-R^2 0.68 · all key coefficients $p < 0.001$
- Coefficients applied to county features and rescaled so county μ's sum to the calibrated state μ
- Combined risk = pathway μ × climate fraction
- 80% credible intervals propagated to every cell

VALIDATE

- Out-of-sample backtest (3 held-out windows, no leakage)
- ROI translation with transparent economic assumptions

One canonical join — every analysis reads from the same Parquet

State labels, county delivery — fit where labels exist, deliver where decisions live

Forecast features ≠ training labels — features are observable; labels are supervised

> **178 MB of raw inputs collapse to 3 MB of analysis-ready outputs. The dashboard reads Parquets directly, no on-the-fly aggregation — that is what makes it fast.**

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Five Tabs, One Workflow

Priorities	Where should I focus this quarter?
Surveillance	Where do extra inspector-hours catch the most?
Pathways	Which origin countries drive the risk?
Establishment	Which counties could sustain a fly if it arrived?
Model	How confident is the model? What is it missing?

All views default to the **next 90 days** — the operational window. The 12-month outlook is preserved as planning context but visually demoted, because forecast confidence degrades meaningfully past one quarter.

The dashboard is designed for program managers, not data scientists. Every tab surfaces a single decision — no raw model output exposed without interpretation.

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Case Study: Q4 2025 Reallocation — Texas vs. California

64%

Top-10 hit rate

3.4× better than persistence baseline

52%

Jul–Sep 2025 hit rate

12 of 23 events

75%

Oct–Dec 2025 hit rate

15 of 20 events

80%

Full CY 2024 hit rate

4 of 5 events

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Out-of-Sample Backtest Results

64% top-10 hit rate — approximately 3.4× the persistence baseline and 50× random chance. Detection events were held out by window; the model was refit on the remainder with no data leakage.

Window	Events	PESTCAST Top-10	Persistence Top-10	Random	PESTCAST / Persistence
Jul–Sep 2025	23	12 (52%)	3 (13%)	1.1%	4.0×
Oct–Dec 2025	20	15 (75%)	5 (25%)	1.1%	3.0×
Full CY 2024	5	4 (80%)	1 (20%)	0.3%	4.0×
TOTAL	**48**	**31 (64%)**	**9 (19%)**	—	**≈3.4×**

Persistence baseline = rank cells by historical detection counts at the same (state, month, species). PESTCAST clears this bar by 3–4× across all test windows.

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Annualized Impact: 731× ROI

$95M

Averted Damage / Year

Annualized estimate based on out-of-sample backtest detections and published eradication cost assumptions.

731×

Return on Investment

Against $130K annual operating cost — robust to ±2× movement in any single economic assumption.

$2.5M

Averted Cost / Detection

5% establishment probability × $50M average eradication cost (CA Medfly: $50–150M historically).

Per-Window Backtest Results

- Jul–Sep 2025: +9 detections vs. persistence → $22.5M averted
- Oct–Dec 2025: +10 detections → $25M averted
- Full CY 2024: +3 detections → $7.5M averted

Economic Assumptions

- Probability undetected → establishes: 5%
- Average eradication cost: $50M
- Tool annual operating cost: $130K
- All assumptions are transparent and override-able in the dashboard

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Where to use PESTCAST

Trust PESTCAST for:

- Ranking which counties, months, and species deserve priority
- Defending budget allocations with quantitative backing
- Surfacing under-watched cells (e.g., Nevada 90-day operational hot spots)
- Communicating risk patterns to leadership in defensible numbers

Do not trust PESTCAST for:

- Exact detection counts — credible intervals are wide on n=52 validation
- Field-level GPS trap placement decisions
- CA Central Valley irrigation-driven establishment
- Connecting-flight attribution (T-100 is bilateral only)
- Unprecedented events — model assumes year-over-year structural stability

> *"Decision-support quality, not prediction-tournament quality. Honest about the 90-day operational horizon and the limits beyond it."*

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